



Forecasting Models for Total Crude Palm Oil Productions in Thailand

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ABSTRACT

This research aims to find a suitable forecasting model for Thailand's total crude palm oil production. The monthly total crude palm oil production in Thailand was gathered from the Office of Agricultural Economics, Ministry of Agriculture, and cooperatives from January 2010 to December 2022. The data were divided into two sets. The first set, from January 2010 to December 2021, was used for constructing and selecting the forecasting models. The second one, from January 2022 to December 2022, was used to compute the accuracy of the forecasting model. Since the total crude palm oil production has trend and seasonal variation, the research used the Holt-Winters method with different initial settings for trend and seasonal influence, the Bagging Holt-Winters method, and the Box-Jenkins method to construct the forecasting models. The minimum mean square error (MSE) and residuals have normal distributions used to select the appropriate forecasting model, and the mean absolute percentage error (MAPE) was used to compute the efficiency of the forecasting model.

According to the three forecasting methods results, the Box-Jenkins method was suitable for forecasting Thailand's total crude palm oil production. The $ARIMA(2,1,2)(0,1,1)_{12}$ model was the best model for predicting Thailand's total crude palm oil production and yielded the $MAPE = 13.49\%$

CCS CONCEPTS

• Applied computing; • Operations Research; • Forecasting;

KEYWORDS

Crude palm oil production, Holt-Winters method, Bagging Holt-Winters method, Box-Jenkins method

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1 INTRODUCTION

Crude palm oil extracts from squeezing palm pulp look like a reddish-brown liquid. Crude palm oil is a raw material in many industries [1]. In general, the uses of crude palm oil are divided into two types: used for consumption, such as snacks, instant noodles, sweetened condensed milk, and margarine, and used as alternative energy, such as B7 diesel, B10 diesel, and B20 diesel.

Thailand's palm oil production is likely to increase according to the policies of the Ministry of Agriculture and Cooperatives. In addition, the 20-year oil palm reform strategy 2016-2036 has promoted and supported the development of the oil palm industry and the entire supply chain. They are supporting renewable energy by using palm oil as an ingredient. For this reason, relevant agencies must have accurate forecast values for making decisions and creating strategic plans for managing Thailand's total crude palm oil to determine an appropriate strategy [2].

The Holt-Winters method suits time series with linear trends and seasonal influences. It is a simple and popular method used in time series forecasting. The initialization of the trend and the seasonal influence affect the performance of the Holt-Winters method [3]. Suppalakpanya et al. [4] created models for crude palm oil production and prices in Thailand using the Holt-Winters and Extended Additive Holt-Winters (EAHW) methods, which set 4 different initial values for trend and seasonal factors. The results found that the different initializations in the two forecasting methods yield significantly different MAPE values. Wongoutong [5] proposes a new initialization for Holt-Winters with a multiplicative model compared with the original and Hansun setup. The results showed that the new configuration settings produced the best results for all 10 data sets.

Bergmeir et al. [6] presented a technique that combines bootstrap aggregation with exponential smoothing methods, and the results showed that it significantly improved forecasts for monthly data. Dantas et al. [7] combined the Holt-Winters method with Bootstrap aggregating, called Bagging Holt-Winters, to predict air transportation demand. The Bagging Holt-Winters gives a more accurate prediction for 13 of 14 monthly time series than other methods. *Bagging* is a bootstrap sampling or replacement sampling method that creates and aggregates many bootstrap samples into one final model from all the bootstrap samples. Banditvilai and Kuharattanachai [8] employed the Holt-Winters method with different initial settings, the Bagging Holt-Winters and Box-Jenkins method, to model and forecast the number of Hat Yai International Airport Passengers. The result shows that the Bagging Holt-Winters model gives more accurate results than other methods.

Box-Jenkins method is famous for its high accuracy in short-term forecasts. Amnah et al. [9] used the Box-Jenkins method to

forecast crude palm oil production in Malaysia and found that the technique provided reliable and accurate long-term forecasts. Alam [10] used ANN and ARIMA models to forecast total imports and exports in Saudi Arabia. The results show that ANN and ARIMA are suitable for predicting Saudi Arabia's total annual imports and exports.

Therefore, this research employs the Holt-Winters Exponential Smoothing method with different settings, the Bagging Holt-Winters method, and Box-Jenkins in forecasting Thailand's total crude palm oil production.

2 DATA COLLECTION AND METHODOLOGY

Monthly crude palm oil production data were collected from the Office of Agricultural Economics, Ministry of Agriculture and Cooperatives. They were collected from January 2010 to December 2022 and divided into two sets. The first set, from January 2010 to December 2021, was used to build the model, and the lowest value of Root Mean Square Error (RMSE) and residuals have a normal distribution used in model selection. The second set, from January 2022 to December 2022, measures performance using Mean Absolute Percentage Error (MAPE).

2.1 Holt-Winters method with different initial values

The Holt-Winters exponential smoothing method suits linear trends and seasonal variation. There are three smoothing equations: level, trend, and seasonal variation. The time series characteristics determine which model to use. When the seasonal component is constant, an additive model is used. When the size of the seasonal component is proportional to the trend level, a multiplicative model is used [11].

2.1.1 Additive Holt-Winters Model. The additive Holt-Winters model is described by equation 1), which has a linear trend with a fixed growth rate (β_1) and a fixed seasonal pattern (S_t) with constant variation.

$$Y_t = (\beta_0 + \beta_1 t) + S_t + \varepsilon_t \quad (1)$$

The level at period $t-1$ is $T_{t-1} = \beta_0 + \beta_1(t-1)$ and at period t is $T_t = \beta_0 + \beta_1 t$. Hence, the growth rate is β_1 . Y_t, ε_t are the observed data and errors at period t .

The estimated level, growth rate, and seasonal factor at period t ($\hat{T}_t, b_t$, and $\hat{S}_t$) are given by the smoothing equations 2)-(4) [10], where α, γ, δ are smoothing constants between 0 and 1. $\hat{T}_{t-1}$ and b_{t-1} are estimated values for the level and growth rate at period $t-1$, $\hat{S}_{t-L}$ is the seasonal factor estimate at period $t-L$, and L denotes the number of seasons per year.

$$\hat{T}_t = \hat{T}_{t-1} + b_{t-1} + \alpha e_t \quad (2)$$

$$b_t = b_{t-1} + \alpha \gamma e_t \quad (3)$$

$$\hat{S}_t = \hat{S}_{t-L} + (1 - \alpha) \delta e_t \quad (4)$$

2.1.2 Multiplicative Holt-Winters Model. The multiplicative Holt-Winters model is described by equation 5), which has a linear trend with a fixed growth rate (β_1) and a fixed seasonal pattern (S_t) with increasing variation,

$$Y_t = (\beta_0 + \beta_1 t) \times S_t \times \varepsilon_t \quad (5)$$

The smoothing equations for the multiplicative Holt-Winters model are given by equations 6)-(8).

$$\hat{T}_t = \hat{T}_{t-1} + b_{t-1} + \frac{\alpha e_t}{\hat{S}_{t-L}} \quad (6)$$

$$b_t = b_{t-1} + \frac{\alpha \gamma e_t}{\hat{S}_{t-L}} \quad (7)$$

$$\hat{S}_t = \hat{S}_{t-L} + \frac{(1 - \alpha) \delta e_t}{T_t} \quad (8)$$

The Holt-Winters method requires starting values, and this research separated starting values into two methods. The first method used the first two years of data to calculate the level component and growth rate, defined in Table 1. Equations 9)-(10) explained the seasonal factors for the additive and multiplicative models of patterns 1-5, respectively.

$$\hat{S}_i = Y_i - \hat{T}_0 \quad ; i = 1, \dots, L \quad (9)$$

$$\hat{S}_i = \frac{Y_i}{\hat{T}_0} \quad ; i = 1, \dots, L \quad (10)$$

The second one employed the Decomposition method, which spent the data from the first set 2-9 years ($T=2, \dots, 9$) calculating trend and seasonal variation ($b_0, b_1, \hat{S}_t$) by employing the ratio to moving average method to decompose. Then it is applied $\hat{T}_t = b_0 + b_1 t$, b_1 and $\hat{S}_t$ for level, growth rate, and seasonal factors.

This research estimated the smoothing parameters (α, γ, δ) and minimizing the root mean square error (RMSE) using the Solver module in Microsoft Excel.

2.2 Bagging Holt-Winters method

The Bagging Holt-Winters method has four steps.

Step 1: Separate time series into trend (T_t) and seasonal factors (S_t) using the ratio to moving average method. The remainder (e_t) can be calculated from Equation

$$e_t = Y_t - \hat{Y}_t \quad (11)$$

where Y_t is the observed data, and $\hat{Y}_t$ is the forecast value, defined by equations 12)-(13).

$$\text{for additive model } \hat{Y}_t = T_t + S_t \quad (12)$$

$$\text{for multiplicative model } \hat{Y}_t = T_t \times S_t \quad (13)$$

Step 2: Mass generation of new time series from random residuals using the Moving Block Bootstrap method with residuals random sampling range of 2-9 periods ($c=2, 3, \dots, 9$). It is randomized 100 times; then, the random residual is combined with the trend and seasonal factor in step 1 to create 100 new time series for each random sampling range (c). An example of residual random sampling range ($c=4$) using the Moving Block Bootstrap method is shown in Figure 1.

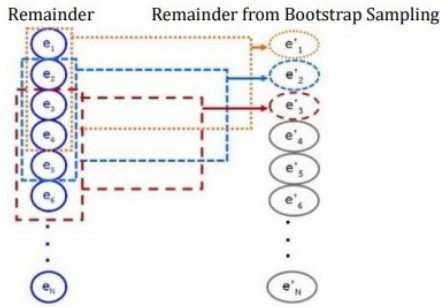
Step 3: This step takes the new time series reconstructed in Step 2 and analyzes them using the Holt-Winters method, with initial values of trends and seasonal influences according to patterns 1, 3, and 4.

Step 4: This step combines the forecast values of 100 new time series into a single set using the median, which is good even though the data has outliers.

This research employed Visual Basic for Application to analyze the Bagging Holt-Winters method.

Table 1: The level and growth rate for five patterns

Pattern	Level component	Growth rate	Reference
1	$\hat{T}_0 = \frac{Y_1 + Y_2 + \dots + Y_L}{L}$	$b_L = \frac{1}{L} \sum_{i=1}^L (Y_{i+L} - Y_i)$	[13, 14]
2	$\hat{T}_0 = \frac{Y_1 + Y_2 + \dots + Y_L}{L}$	$b_L = Y_2 - Y_1$	[4, 15]
3	$\hat{T}_0 = \frac{Y_1 + Y_2 + \dots + Y_L}{L}$	$b_L = \frac{(Y_2 - Y_1) + (Y_3 - Y_2) + (Y_4 - Y_3)}{3}$	[4, 15]
4	$\hat{T}_0 = \frac{Y_1 + Y_2 + \dots + Y_L}{L}$	$b_L = \frac{Y_L - Y_1}{L-1}$	[4, 13-16]
5	$\hat{T}_0 = \frac{Y_1 + Y_2 + \dots + Y_L}{L}$	$b_L = 0$	[4, 8, 13-16]

**Figure 1: Remainder from Bootstrap Sampling when c=4**

2.3 Box-Jenkins method

The Box-Jenkins method was developed in 1974 and is widely used in time series forecasting. The Box-Jenkins method defines a model based on the characteristics of Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF). Box-Jenkins method consists of 4 steps as follows [17]

Step 1: Determine the appropriate expected model for the time series.

Step 2: Use historical data to estimate parameters in the model.

Step 3: Check if the model in Step 1 fits the time series. If unsuitable, a new adjusted model is suggested and returned to step 2.

Step 4: The suitable model is used to predict future values. In cases where there are several suitable models

Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used to select a single optimal model. AIC and BIC are defined as follows [18]:

$$AIC = 2 \ln L + 2k \quad \text{and} \quad BIC = 2 \ln L + \ln(n)k$$

Where $L = RSS/n$ is the sum of squared residuals, n is the number of residuals calculated from the time series, and k is the number of parameters estimated in the model. The most appropriate model gives the least AIC and BIC.

The Box-Jenkins model $ARIMA(x, d, z)(X, D, Z)_L$ is defined as follows [17]:

$$\phi_X(B)\Phi_X(B^L)Z_t = \theta_0 + \theta_z(B)\Theta_Z(B^L)\varepsilon_t \quad (14)$$

Where

$$Z_t = (1 - B)^d (1 - B^L)^D$$

$$\phi_X(B) = (1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_x B^x)$$

$$\theta_z(B) = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_z B^z)$$

$$\Phi_X(B^L) = (1 - \Phi_1 B^L - \Phi_2 B^{2L} - \dots - \Phi_X B^{XL})$$

$$\Theta_Z(B^L) = (1 - \Theta_1 B^L - \Theta_2 B^{2L} - \dots - \Theta_Z B^{ZL})$$

B is the backward shift operator.

Y_t is the observed data at period t .

θ_0 is a constant.

$\phi_X(B)$ is the non-seasonal autoregressive model of order x .

$\theta_z(B)$ is the non-seasonal moving average model of order z .

$\Phi_X(B^L)$ is the seasonal autoregressive model of order X .

$\Theta_Z(B^L)$ is the seasonal moving average model of order Z .

ε_t is the error at period t and has a normal distribution, with a mean equal to zero and constant variance and statistically independent.

d is the number of regular differences.

D is the number of seasonal differences.

This research employed Minitab 21.0 to analyze the Box-Jenkins model.

3 MODEL SELECTION CRITERION

The smallest root mean square error (RMSE) and residuals have a normal distribution employed in selecting the predicting model. The mean absolute percentage error (MAPE) measures the model's accuracy.

$$RMSE = \sqrt{\frac{\sum_{t=1}^n e_t^2}{n}}$$

$$MAPE = \frac{\sum_{t=1}^n \left| \frac{e_t}{Y_t} \right|}{n} \times 100$$

4 RESULTS

The first set of data from January 2010 to December 2021 is employed to build the Holt-Winters models with different initial settings, the Bagging Holt-Winters and Box-Jenkins models.

4.1 The Holt-Winters method

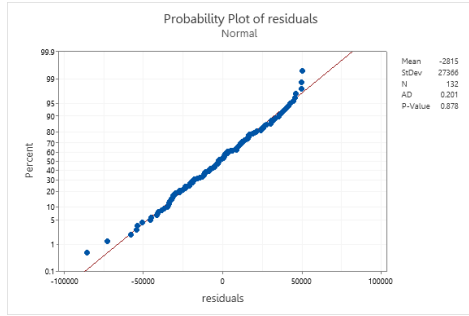
Tables 2 and 3 show the Holt-Winters models with results from different initial settings.

*residuals have a normal distribution

From Table 2, the Holt-Winters models with initial settings from pattern 5 obtained the minimum RMSE for additive models. However, the residuals from the additive model from pattern five do not have a normal distribution. From the Anderson-Darling test for normal distribution at a significant level of 0.05, since the p-value of 0.878 is greater than $\alpha = 0.05$, the additive Holt-Winters model

Table 2: RMSE of Holt-Winters models with five different initial values

Pattern	1	2	3	4	5
Additive model	*26,831.80	28,620.24	*27,319.83	*27,104.20	26,697.27
Multiplicative model	39,525.60	40,161.49	39,870.20	39,947.18	39,469.59

**Figure 2: Test of Normal distribution for residuals from the additive Holt-Winters model with initial setting from pattern 1**

residuals with initial settings pattern 1 have a normal distribution as shown in Figure 2. Therefore, the additive Holt-Winters model with initial settings from pattern one gives the smallest RMSE (26,831.80) and residuals a normal distribution.

*residuals have a normal distribution

From Table 3, the multiplicative Holt-Winters model with initial values from the 6-year decomposition method yields a minimum RMSE (28,733.92), and residuals have a normal distribution.

4.2 The Bagging Holt-Winters method

Since the residuals obtained from the initial settings of patterns 1, 3, and 4 have a normal distribution. This research set the initial settings for the Bagging Holt-Winters method to be pattern 1, 3, and 4, with the random sampling range ($c=2, 3, \dots, 9$). Tables 4, 5, and 6 show the results of the Bagging Holt-Winters models with initial settings from patterns 1, 3, and 4, respectively. From Table 4, the RMSE of additive and multiplicative models increases when the random sampling range increases from $c=2$ to $c=9$. The additive Bagging Holt-Winters model with random sampling range ($c=2$) gave the minimum RMSE (23,408.24). However, In the Bagging Holt-Winters method, initialized with pattern one, there is no random sampling range in which residuals have a normal distribution.

*residuals have a normal distribution

Table 3: RMSE of Holt-Winters models with initial values from the Decomposition method.

Number of years (T)	2	3	4	5
Additive model	25,110.77	28,231.06	*32,426.17	*29,590.45
Multiplicative model	34,045.51	29,807.03	*34,071.43	*29,346.95
Number of years (T)	6	7	8	9
Additive model	27,336.92	27,461.19	*31,924.72	*30,233.27
Multiplicative model	*28,733.92	29,095.07	34,185.42	30,705.41

*residuals have a normal distribution

From Table 5, the RMSE of additive and multiplicative models increases when the random sampling range increases from $c=3$ to $c=9$. The additive Bagging Holt-Winters model with initial setting pattern three and random sampling range ($c=4$) gave the minimum RMSE (31,135.85), and residuals have a normal distribution.

*residuals have a normal distribution

From Table 6, the additive Bagging Holt-Winters model with initial setting pattern four and random sampling range ($c=2$) gave the minimum RMSE (33,808.99), and residuals have a normal distribution

4.3 Box-Jenkins method

From Figure 3, the total crude palm oil production is nonstationary since it has a linear trend and seasonal variation. To make it stationary one regular difference ($d=1$) and one seasonal difference ($D=1$) are taken. Based on the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF), as shown in Figures 4 and 5, ACF and PACF are decreasing rapidly, suggests ARIMA(1,1,1). Additionally, ACF showed a significant difference at lag 12, and PACF at lag 12, 24, 36, 48, ... is exponentially decreasing, suggesting a seasonal (0,1,1). However, ARIMA(1,1,1) (0,1,1)₁₂ model does not pass the diagnostic check. Therefore, the newly identified model is ARIMA(2,1,2) (0,1,1)₁₂.

Table 7 shows all parameters ($\phi_1, \phi_2, \theta_1, \theta_2, \Theta_{12}$) from ARIMA(2,1,2)(0,1,1)₁₂ model is statistically significant from zero since the p-value is less than the significance level of 0.05. The modified Box-Pierce test shows that residuals from the model are statistically independent since the p-value is greater than 0.05, and residuals have a normal distribution, as shown in Figure 6. From the Anderson-Darlings test in Figure 6, ARIMA(2,1,2)(0,1,1)₁₂ model's residuals have a normal distribution because the p-value of 0.802 is greater than the significant level of 0.05. Therefore, the model ARIMA(2,1,2)(0,1,1)₁₂ fits with Thailand's total crude palm oil production. In addition, ARIMA(2,1,2)(1,0,1)₁₂ also passes the diagnostic checking. Table 8 shows all Box-Jenkins models with RMSE, AIC, and BIC.

From Table 8, the ARIMA(2,1,2)(0,1,1)₁₂ model gives the minimum AIC and BIC, the ARIMA(2,1,2)(0,1,1)₁₂ is the optimum model

Table 4: RMSE of the Bagging Holt-Winters models with initial settings from pattern 1.

random sampling range (c)	2	3	4	5
Additive model	23,408.24	26,531.30	29,766.27	32,206.16
Multiplicative model	35,365.31	35,191.01	37,998.34	39,684.77
random sampling range (c)	6	7	8	9
Additive model	35,037.19	35,679.44	37,055.76	37,787.77
Multiplicative model	40,948.09	41,972.17	40,838.30	40,751.49

Table 5: RMSE of the Bagging Holt-Winters models with initial settings from pattern 3.

random sampling range (c)	2	3	4	5
Additive model	24,169.37	27,275.43	*31,135.85	*34,858.69
Multiplicative model	37,471.96	36,052.70	39,086.28	42,109.05
random sampling range (c)	6	7	8	9
Additive model	*37,476.87	*39,476.87	*40,866.05	*42,441.83
Multiplicative model	43,818.54	43,850.32	44,454.29	44,047.61

Table 6: RMSE of the Bagging Holt-Winters models with initial settings from pattern 4.

random sampling range (c)	2	3	4	5
Additive model	23,581.79	26,666.42	29,390.61	35,909.28
Multiplicative model	*48,073.51	*44,518.31	*39,011.84	37,343.85
random sampling range (c)	6	7	8	9
Additive model	*33,808.99	*35,417.63	*35,909.28	*36,973.28
Multiplicative model	39,602.93	*39,252.66	*40,221.09	*40,462.28

Table 7: Minitab output of ARIMA(2,1,2) (0,1,1)₁₂

Type	Coef	SE Coef	t	p-value
AR1($\hat{\phi}_1$)	1.5362	0.0643	23.89	0.000
AR2($\hat{\phi}_2$)	-0.7567	0.0707	-10.71	0.000
MA1($\hat{\theta}_1$)	1.2171	0.0447	27.25	0.000
MA2($\hat{\theta}_2$)	-0.2878	0.0614	-4.68	0.000
MA12($\hat{\Theta}_{12}$)	0.8519	0.0755	11.29	0.000
Modified Box-Pierce(Box-Ljung)Chi-Square statistic				
Lag	12	24	36	48
Chi-Square	9.69	19.25	39.90	47.77
DF	7	19	31	43
p-value	0.207	0.441	0.131	0.285

for Thailand's total crude palm oil production. Figure 7 shows actual data, fits, and forecast value for one year ahead.

4.4 MODEL SELECTION AND PERFORMANCE MEASURE

From Table 9, the additive Holt-Winters model with initial values from pattern one gives the minimum MAPE of 11.74%. Since the

Table 8: RMSE, AIC, and BIC of Box-Jenkins models

Box-Jenkins Model	RMSE	AIC	BIC
ARIMA(2,1,2)(0,1,1) ₁₂	17,763.5	49.14	63.52
ARIMA(2,1,2)(1,0,1) ₁₂	17,435.1	51.06	68.84

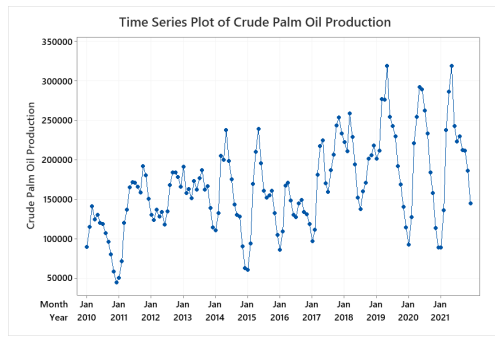


Figure 3: Thailand's total crude palm oil production from January 2010 to December 2021.

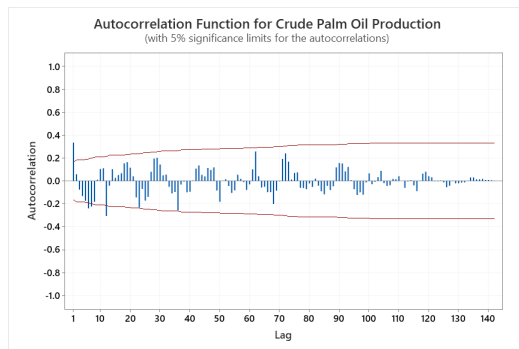


Figure 4: : ACF of crude palm oil production with one regular difference and one seasonal difference.

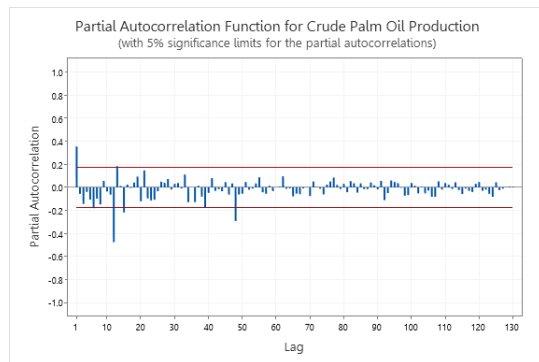


Figure 5: PACF of crude palm oil production with one regular difference and one seasonal difference.

criteria for choosing the optimal model is minimum RMSE and residuals have a normal distribution, the Box-Jenkins method is the most appropriate for predicting Thailand's total crude palm oil production, giving an RMSE of 17,763.55. This is followed by the Holt-Winters method with an initial settings pattern 1, which provides an RMSE of 26831.80. This research confirms that the Box-Jenkins method is a powerful tool for forecasting. This research shows that the different initial values can increase the efficiency

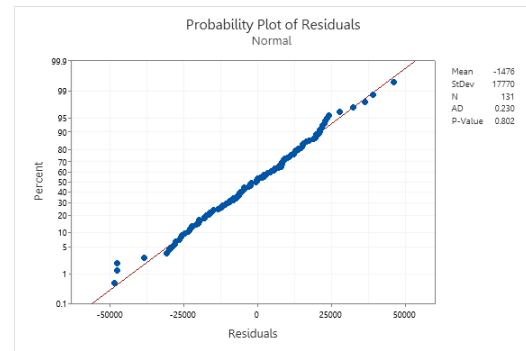


Figure 6: Test of Normal distribution for residuals from ARIMA(2,1,2)(0,1,1)₁₂ model

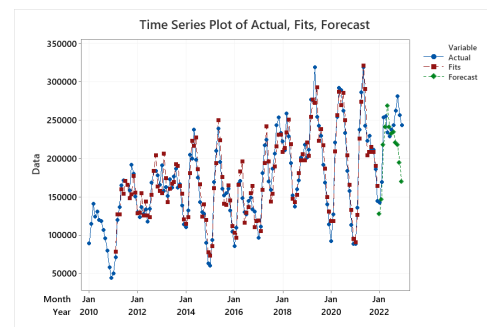


Figure 7: Actual, Fits, and Forecast by the Box-Jenkins model

of the Holt-Winters method if the time series have linear trends and seasonal variation, similar to Thailand's total crude palm oil production.

4.5 CONCLUSION

This research presents three forecasting methods: the Holt-Winters method with different initial settings, the Bagging Holt-Winters method, and the Box-Jenkins method to model Thailand's total crude palm oil production. The results showed that the Box-Jenkins method gives the appropriate model for total crude palm oil production in Thailand and yields the MAPE 13.49%. The Holt-Winters method with different initial values is more straightforward and requires less data to estimate than the Box-Jenkins method and Bagging Holt-Winters method. However, its performance outperformed the Bagging Holt-Winters method. The increased complexity of the Bagging Holt-Winters method does not guarantee better results. This research shows that the more straightforward Holt-Winters method with different initial values can give higher accuracy in some cases. When the researcher models time series data with linear trend and seasonal factors, the Box-Jenkins method and the Holt-Winters method with different initial values are good alternatives.

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Table 9: The RMSE and MAPE from Holt-Winters, Bagging Holt-Winters, and Box-Jenkins method

Forecasting model	RMSE	MAPE
The additive Holt-Winters with initial settings from pattern one	26,831.80	11.74%
The additive Holt-Winters with the initial settings from the decomposition method (T=6)	28,733.92	14.82%
The additive Bagging Holt-Winters model with initial settings pattern three and random sampling (c=4)	31,135.85	18.42%
The additive Bagging Holt-Winters model with initial settings pattern four and random sampling (c=6)	33,808.99	16.49%
The ARIMA(2,1,2) (0,1,1) ₁₂	17,763.55	13.49%

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